Performance Metrics in Computer Systems

1. Response Time

Definition: Time taken from start to finish of a specific task.

Also called *execution time*.

Includes:

- CPU processing time
- Memory access time
- Disk I/O
- Operating system overhead

> **Note**
>
> Useful for evaluating how fast a system completes individual tasks.

2. Throughput

Definition: Number of tasks completed per unit time.

Examples:

- Jobs per hour
- Requests per second

> **Usefulness**
>
> Important in multitasking environments and server systems.

3. Relationship: Response Time vs Throughput

In general:

$$\text{Lower Response Time} \quad \Rightarrow \quad \text{Higher Throughput}$$

- Faster task completion allows more tasks to be handled in the same time.
- However, real systems have limits due to CPU, memory, I/O, etc.

4. Performance and Execution Time

Formula:

$$\text{Performance}_x = \frac{1}{\text{Execution Time}_x}$$

Implication:

- Shorter execution time $\Rightarrow$ higher performance
- Faster systems = lower execution time

5. Comparing Two Computers (X vs Y)

If:

$$\text{Performance}_x > \text{Performance}_y$$

Then:

$$\frac{1}{\text{Execution Time}_x} > \frac{1}{\text{Execution Time}_y} \quad \Rightarrow \quad \text{Execution Time}_x < \text{Execution Time}_y$$

Conclusion: Computer X completes tasks faster than Y.

6. Speedup (n times faster)

If computer X is n times faster than Y:

$$\frac{\text{Performance}_x}{\text{Performance}_y} = n \quad \text{OR} \quad \frac{\text{Execution Time}_y}{\text{Execution Time}_x} = n$$

Example:

- Y takes 10 seconds
- X is $2\times$ faster
- Then: X takes $10/2 = 5$ seconds

7. Summary Table

Term	**Definition**	**Goal**
Response Time	Time to finish one task	Minimize
Throughput	Number of tasks per unit time	Maximize
Performance	Inverse of execution time	Higher is better
Speedup	Ratio of execution times between systems	Shows how much faster